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Multi-Hop-Biomedical-Reasoning

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SUPERVISOR CERTIFICATION

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“Multi-Hop Biomedical Reasoning ”
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الملخص

تواجه النماذج اللغوية الكبيرة تحدياً جوهرياً في الاستدلال الطبي متعدد المراحل، حيث تعجز عن ربط الأدلة المتناثرة بدقة وموثوقية. تكتسب هذه المسألة أهمية بالغة نظراً لحساسية المجال الطبي وتأثيره المباشر على سلامة المرضى، مما يفرض معايير صارمة للدقة وقابلية التدقيق. إلا أن التطبيقات الحالية تعاني من تضخيم الأخطاء والهلوسة المتسلسلة بسبب الطبيعة الاحتمالية للتوليد، بالإضافة إلى "فجوة التركيب" التي تعيق دمج الحقائق المنفصلة منطقياً. كما أن الحلول عالية الدقة تعتمد حالياً على نماذج احتكارية وسحابية مكلفة، مما يرفع الحواجز الحسابية ويعيق نشرها في البيئات محدودة الموارد، ويستدعي تطوير بديل عملي يوازن بين الدقة والكفاءة والتكلفة المنخفضة.

يقدم البحث نظاماً محلياً (Local-First) يعتمد على نموذج Qwen3.5-9B مفتوح المصدر لمعالجة الأسئلة المعقدة دون واجهات خارجية أو ضبط مسبق. تكمن قيمته المضافة في دمج ثلاث مكونات مبررة وظيفياً وهي استرجاع هجين يجمع التضمين الدلالي (MedCPT) مع فرز معجمي مرجح، لضمان التقاط السياق الدلالي وتعزيز الدقة عبر مصطلحات حاسمة وتجنب تسمم السياق. ووحدة توسيع هيكلية تصنف المصطلحات طبياً وتمنحها أوزاناً تعكس أولويتها في التنبؤ بالتفاعلات، مما يقلل الضوضاء ويركز الاسترجاع. وأخيراً، هندسة أوامر موجهة تفرض قيوداً صارمة على المخرجات وتوفر أمثلة استدلالية، لمنع انحراف النموذج ومحاكاة منطق الخبراء دون إعادة تدريب. يضمن هذا التصميم شفافية عالية، وتكلفة تشغيل شبه معدومة، وآلية تحقق محلية تقلل الاعتماد على المعرفة الداخلية غير الموثوقة.

أظهرت أكثر من خمسين تجربة أن النظام حقق دقة تطابق تام (Exact Match) بنسبة 33.3% على مجموعة MedHop (342 سؤالاً)، محققاً تحسناً +16.9% فوق خط الأساس، ومتفوقاً على النماذج المضبوطة مثل CaresAI (18.6%) ومنافساً للحلول المحلية الأقوى مثل Orekhovich (43.9%). كشف تحليل الأخطاء أن 60% من الإخفاقات ناتجة عن اختيار مرشح خاطئ لغياب الأدلة القاطعة في النصوص، بينما انخفضت الهلوسة إلى 7.9%. وأثبتت النتائج أن عدد مستندات K=3 هو الأمثل لتجنب تشتت الانتباه، وأن السقف النظري للاسترجاع المثالي (Oracle) بلغ 76.9%، مما يكشف فجوة أداء قدرها 43.6% تعزى أساساً إلى ضوضاء المستندات الداعمة وافتقارها لروابط سببية صريحة. يخلص البحث إلى أن جودة البيانات ووضوح الروابط المنطقية عامل حاسم لا يقل أهمية عن حجم النموذج، وأن دمج الاسترجاع الهجين مع التوجيه الصارم للأوامر يمثل مساراً عملياً وقابلاً للنشر الفوري لأنظمة استدلال طبية موثوقة ومنخفضة التكلفة.

الكلمات مفتاحية: الاستدلال الطبي متعدد المراحل، النماذج اللغوية المحلية، الاسترجاع الهجين الموزون، هندسة الأوامر الموجهة، أنظمة الاستدلال منخفضة التكلفة.

Abstract

Large language models face a fundamental challenge in multi-hop medical reasoning, as they fail to connect scattered evidence with accuracy and reliability. This issue holds critical importance due to the sensitivity of the medical field and its direct impact on patient safety, which imposes strict standards for accuracy and auditability. Current applications suffer from error amplification and cascading hallucinations due to the probabilistic nature of text generation, alongside a compositionality gap that hinders the logical integration of separate facts. High-precision solutions currently rely on expensive proprietary and cloud-based models, which raise computational barriers and hinder deployment in resource-constrained environments, necessitating the development of a practical alternative that balances accuracy, efficiency, and low cost.

The research presents a local-first system built on the open-source Qwen3.5-9B model to process complex questions without external interfaces or prior fine-tuning. Its added value lies in integrating three functionally justified components. The system employs hybrid retrieval that combines semantic embedding through MedCPT with weighted lexical ranking to capture semantic context, enhance accuracy through critical terms, and prevent context poisoning. A structural expansion module classifies medical terms and assigns weights that reflect their priority in predicting interactions, which reduces noise and focuses retrieval. Guided prompt engineering imposes strict output constraints and provides reasoning examples to prevent model deviation and simulate expert logic without retraining. This design ensures high transparency, near-zero operational cost, and a local verification mechanism that reduces reliance on unreliable internal model knowledge.

More than fifty experiments demonstrated that the system achieved an Exact Match accuracy of 33.3 percent on the MedHop dataset of 342 questions, delivering a 16.9 percent improvement over the baseline, outperforming fine-tuned models such as CaresAI at 18.6 percent, and competing with stronger local solutions like Orekhovich at 43.9 percent. Error analysis revealed that 60 percent of failures stem from selecting an incorrect candidate due to the absence of decisive evidence in the texts, while hallucination decreased to 7.9 percent. The results confirmed that K equals 3 documents represents the optimal threshold to prevent attention dispersion, and that the theoretical ceiling for oracle retrieval reached 76.9 percent, exposing a performance gap of 43.6 percent that primarily originates from noise in supporting documents and their lack of explicit causal links. The research concludes that data quality and clarity of logical links constitute a decisive factor no less important than model size, and that integrating hybrid retrieval with strict prompt guidance establishes a practical and immediately deployable pathway for reliable, low-cost medical reasoning systems.

Keywords: Multi-hop medical reasoning, local language models, weighted hybrid retrieval, guided prompt engineering, low-cost reasoning systems.

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List of Terms and Abbreviations

Terminology standards: IEEE AI Glossary | ACL/Stanford NLP | BioNLP peer-reviewed literature

Table 2: List of Terms and Abbreviations

Term	Definition	Abbrev.
Large Language Model	A neural language model trained on massive text corpora, capable of generating and understanding natural language	LLM
Retrieval-Augmented Generation	A framework that enhances LLM generation by retrieving relevant external documents at inference time	RAG
Exact Match	Evaluation metric that awards a score of 1 only when the predicted answer is identical to the gold answer	EM
Chain-of-Thought	A prompting technique that guides the model to generate intermediate reasoning steps before the final answer	CoT
Multi-Hop Reasoning	An inference process that requires combining evidence from two or more separate sources to reach a conclusion	—
Hallucination	The generation of factually incorrect or fabricated content not supported by the input or retrieved context	—
Fine-Tuning	Supervised training of a pre-trained model on a task-specific dataset to adapt its behavior	—
Prompt Engineering	The systematic design of input instructions to elicit specific and accurate outputs from a language model	—
Few-Shot Prompting	A prompting strategy that provides a small number of labeled examples within the input to guide model behavior	—
Semantic Retrieval	Document retrieval based on semantic similarity using dense vector representations	—
Lexical Retrieval	Document retrieval based on keyword or term matching, typically using sparse representations	—

Term	Definition	Abbrev.
Hybrid Retrieval	A retrieval strategy that linearly combines semantic and lexical retrieval scores	—
Semantic Embedding	A dense numerical vector representation of text that encodes semantic meaning	—
Cosine Similarity	A geometric measure of the angle between two vectors, used to quantify semantic similarity	—
Bi-Encoder	A dual-encoder architecture that encodes query and document independently into a shared vector space	—
Cross-Encoder	An architecture that jointly encodes query and document via full self-attention for fine-grained relevance scoring	—
Re-Ranking	A secondary retrieval stage that reorders an initial candidate set using a more precise scoring model	—
Reciprocal Rank Fusion	A score aggregation method that combines ranked lists by summing reciprocals of document ranks	RRF
Ontology	A formal, structured representation of concepts and relationships within a knowledge domain	—
Knowledge Graph	A graph-structured database that encodes entities as nodes and their relationships as directed edges	KG
Named Entity Recognition	The NLP task of identifying and classifying mentions of specific entities (drugs, genes, diseases) in text	NER
Context Window	The maximum number of tokens a language model can process in a single inference pass	—
Temperature (T)	A scalar hyperparameter that controls the entropy of the Softmax output distribution during text generation	T
Softmax Function	A mathematical function that converts a vector of logits into a probability distribution over a vocabulary	—

Term	Definition	Abbrev.
Autoregressive Generation	A generation paradigm in which each token is predicted conditioned on all previously generated tokens	—
Quantization	Model compression technique that reduces weight precision (e.g., from 16-bit to 4-bit) to decrease memory requirements	—
Pipeline	A sequential data processing architecture where each module passes its output to the next	—
Modular Architecture	A system design in which components are independent, reusable, and interchangeable	—
Query Expansion	The process of enriching a search query with semantically related or weighted terms to improve retrieval coverage	—
Compositionality Gap	The failure of a model to combine two separately known facts into a logically correct compound conclusion	—
Parametric Knowledge	Knowledge encoded in a model's weights during pre-training, as opposed to retrieved external knowledge	—
Error Propagation	The accumulation and amplification of errors across successive reasoning steps in a multi-hop pipeline	—
Context Poisoning	Degradation of model performance caused by the inclusion of irrelevant or misleading documents in the context	—
Oracle Retrieval	An idealized retrieval method that selects documents based on prior knowledge of the correct answer; used to measure the performance upper bound	—
Upper Bound	The theoretical maximum achievable performance under ideal retrieval conditions	—
Baseline	A reference system or method against which the performance of the proposed approach is measured	—

Term	Definition	Abbrev.
Recall@K	The proportion of relevant items found within the top-K retrieved documents	—
Majority Voting	An ensemble aggregation strategy that selects the answer chosen by the majority of individual systems	—
Ensemble System	A system that combines predictions from multiple models or strategies to improve overall accuracy	—
BM25 / BM25S	A probabilistic lexical retrieval algorithm based on term frequency and inverse document frequency	BM25
MedCPT	A biomedical dense retrieval model trained on clinical query-article pairs from PubMed using contrastive learning	MedCPT
MedHop Dataset	A multi-hop biomedical question answering benchmark developed for the BioCreative IX Shared Task	—
DrugBank	A comprehensive online database combining detailed drug data with drug target information	—
Hetionet	An integrative biomedical knowledge network encoding relationships between diseases, genes, drugs, and pathways	—
Qwen3.5-9B	A 9-billion-parameter open-source language model by Alibaba, used in 4-bit quantized form in this research	—
Ollama	An open-source local server framework for hosting and querying quantized large language models	—
Application Programming Interface	A software interface that allows one application to interact with another through predefined endpoints	API
Concept-Level Score	An evaluation metric that measures semantic overlap between the predicted and reference answers using MedCPT embeddings	—

1 Introduction

1.1 Introduction

Medical research is inherently sensitive due to its direct impact on human health, making absolute accuracy a prerequisite for any intelligent system operating in this field. The phenomenon of hallucination in LLMs emerges as the most significant barrier, particularly during multi-step inference, where errors can accumulate at each stage to produce cascading hallucinations or compounding reasoning errors, ultimately yielding outputs that deviate substantially from factual ground truth [1]. While large-scale monolithic models or computationally intensive LLMs have been proposed as solutions to this challenge, such approaches entail considerable financial cost and demand extensive computational resources, rendering them unsuitable for academic settings and resource-constrained environments [2].

This thesis seeks to provide a practical, intermediate solution that integrates parameter-efficient models or mid-scale models with computationally intensive pipelines. Specifically, it aims to develop a lightweight, interpretable architecture that ensures each reasoning hop is grounded in retrieved, evidence-based information derived from structured medical facts. By combining semantic retrieval with structured concept validation against biomedical ontologies, this research aspires to mitigate error propagation and deliver a reliable, resource-efficient inference system.

1.2 Research Problem

The problem with this research is that current language models are unable to implement a correct step-wise reasoning chain when answering complex biomedical QA tasks, even when the individual facts that constitute the answer are available to them. This shortcoming stems from the lack of a lightweight and feasible mechanism for validating and correcting intermediate reasoning steps in real time during the inference process. The reliance on computationally intensive monolithic models or proprietary commercial APIs leads to high computational and financial costs that hinder the adoption of these solutions in academia and medium-sized enterprises. As a result, the main research problem is how to design an accurate, interpretable, and low-cost multi-hop inference pipeline, without the need for intensive retraining or massive computing resources.

1.3 Research Objectives

This project aims to build an integrated medical inference system that combines local document retrieval with symbolic verification. A module is created to determine whether a question requires single-hop or multi-hop processing (Multi-hop Classification). Techniques such as BM25S are then used to extract relevant paragraphs from a local biomedical corpus instead of relying on external interfaces. A concept-level validation layer against biomedical ontologies is applied to detect relationships between medical entities (diseases, genes, drugs) and correct the reasoning trajectory when errors occur. System performance is evaluated using Exact Match (EM) and Concept-Level (MedCPT) metrics on the MedHop dataset.

1.4 Research Significance

The research provides an intermediate solution that combines multi-hop reasoning with mid-scale, parameter-efficient models that can be deployed locally, thereby reducing financial and computational costs. The research focuses on making the inference process transparent and verifiable, mitigating cascading reasoning errors to a minimum, and ensuring highly reliable and interpretable outputs. In addition, the proposed solution avoids the substantial costs associated with relying on proprietary, closed-source commercial models.

1.5 Beneficiaries of the Research

The proposed algorithm in this thesis has been designed to offer a practical solution to the problem of cost and computational complexity; therefore, the primary beneficiaries are all those who face these obstacles. Consequently, it targets a broad range of beneficiaries, starting with entities that suffer from limitations in resources or budget. The proposed algorithm enables students and researchers at universities to understand and use an advanced medical reasoning algorithm without exorbitant costs. Medical laboratories and small clinics benefit from a transparent and interpretable decision-support tool that operates locally without the need for an internet connection. As for developers and the open-source community, the research provides them with a nucleus for developing customized medical solutions for various subdomains. This algorithm constitutes a strategic solution for healthcare institutions in remote or resource-limited areas, thereby contributing to access to reliable medical artificial intelligence technologies.

1.6 Research Structure

This work is organized into six chapters. Chapter 1 provides a general introduction, outlining the research objectives, significance, and methodology. Chapter 2 presents the theoretical framework, covering foundational concepts in natural language processing, multi-hop reasoning, and information retrieval. Chapter 3 reviews the relevant literature and prior work, documenting the scientific gap this research aims to address. Chapter 4 details the architectural design of the proposed pipeline, explaining the implemented algorithms (e.g., BM25S, Chain-of-Thought prompting) and the integration of ontological validation. Chapter 5 presents the experimental results, including both successful iterations and failure analyses. The final chapter summarizes the key findings and outlines directions for future research.

2 Theoretical Framework

2.1 Computational Properties of Language Models and Their Impact on Medical Reliability

Understanding the internal mechanisms and how large language models work is the first and most important step in the process of evaluating their capabilities and advantages, as well as in revealing their inherent limitations that become dire in contexts such as the biomedical field. These models, as opposed to deterministic knowledge rules, are probabilistic systems in nature, and this foundational principle explains their various patterns of failure.

2.1.1 Autoregressive generation generation and error amplification dynamics

The first challenge lies in the autoregressive generation mechanism that forms the core of the text production process in these models. The regressive model predicts the next token based solely on its predecessor, where this sequential process leads to the phenomenon of path dependence, which can be expressed as the probabilistic formula ($P\left(x_t \vee x_{\{<t\}}\right)$). Any simple error in predicting an intermediate token during a complex, multi-step inference process is not isolated, and this error is automatically and blindly integrated into the context on which all future predictions of that session will be based. This phenomenon amplifies error amplification, or hallucination loops, in which the model gradually deviates from the path of truth, generating a linguistically coherent but factually incorrect narrative [3]. The error may begin with an incorrect assumption about the function of a protein, which is later based on a false conclusion about the mechanism of a disease, ultimately resulting in a serious therapeutic recommendation. This theoretical behavior conclusively illustrates the urgent need for an external, periodic verification mechanism, which acts as a gatekeeper capable of breaking this destructive loop before it extends to the final stages of inference [3].

2.1.2 Probability Distribution and Entropy Control

The model does not necessarily choose the next symbol, but uses a probabilistic distribution that covers its entire vocabulary. The essence of the generation process is to apply a function (Softmax) to the logits to produce this distribution. This is where the temperature coefficient (T comes in), which is a very important coefficient that controls the entropy or intensity of this distribution. Lowering the T value below 0.5 makes the distribution Peakier, which greatly favors symbols with higher probabilities, thus producing a more deterministic and consistent output, and raising the T value above 0.8 increases the flattening of the distribution, allowing for the selection of less likely symbols and thus producing a more creative and diverse but less predictable and more hallucinogenic text. For tasks that require high literal accuracy, such as Medical NER extraction and symbolic checks, T will be set at very low levels (0.0 - 0.3) to ensure realistic consistency and avoid inaccurate paraphrasing. In contrast, the temperature can be slightly raised ($T \approx 0.7$) in the query expansion phase to generate semantic synonyms and alternative formulations for the search, enriching the RAG process without compromising the essence [2].

2.1.3 Contextual memory limitations and their impact on longitudinal reasoning

The Context Window represents the maximum number of input and output symbols that a model can process at once, and despite the massive expansion in the size of these windows in modern models, research has shown that performance is not uniform across the entire window. One of the most prominent of these limitations is the phenomenon of Lost in the Middle, as the performance of models in retrieving and using information is significantly impaired when placed in the middle of a long context, compared to information at the beginning or end [4].

This phenomenon poses a significant risk to medical inference, which requires the aggregation of evidence from different parts of the patient's medical history or from multiple research papers. In addition, models suffer from representation collapse in long contexts, where representations of semantically similar entities overlap. The proposed method relies primarily on the Atomic Recursive Decomposition strategy to counteract this limitation, where the complex medical question is broken down into as short and independent inferential units as possible. This ensures that each heuristic leap operates within a small, effective local attention span, keeping relevant evidence in the focus of the model and avoiding the problem of getting lost in the middle of the context, greatly enhancing the accuracy of local retrieval [4].

2.2 The Theoretical Framework of Multi-Stage Reasoning in the Medical Field

The medical field often requires complex, multi-hop inferences to arrive at an answer, a challenge that language models still face. Understanding the nature of these challenges from a theoretical perspective allows us to design more effective strategies to address them.

2.2.1 Compositionality Gap

The formation gap in the context of linguistic models is defined as the systemic failure of the model to logically correlate two or more facts that it knows separately to produce a complex and correct conclusion [4]. The model may have memorized from its training data the relationship (Parkinson's disease - SNCA gene) and the relationship (SNCA protein - drug A), and when asked the compound question: Can drug A be a potential treatment for Parkinson's disease? the model may fail to carry out the simple reasoning process: $A \leftarrow B \wedge B \leftarrow C \therefore A \leftarrow C$ [3]. This failure stems not from a lack of knowledge, but from the model's inability to recall and integrate this knowledge explicitly and goal-oriented at the same time. The adoption of the explicit decomposition model in this research transforms the implicit heuristic path that the model may fail to follow into a series of discrete steps that can be individually tracked and verified. This gap is closed by forcing the system to generate the first evidence, document it, and then explicitly pass it as a starting point for the second jump [5].

2.2.2 Recursive deconstruction and the chain of ideas as a guiding mechanism

Chain-of-Thought (CoT) technology has emerged as an effective mechanism to enable models to process complex tasks, by generating intermediate steps of textual thinking before arriving at the final answer [5]. The technique has proven to be effective, and explained this from a theoretical perspective that CoT increases the expressive power of the transformer to be able to solve dynamic problems that require sequential processes. CoT is not only used to make the thought process interpretable by humans, but it is also activated as a recursive decomposition mechanism in this research [4]. This means that the model turns into a planner whose first task is to analyze the complex root question and segment it into a tree or matrix of atomic sub-queries [1]. Each atomic question is designed to be solved in a single, verifiable heuristic leap [4]. This philosophy strictly isolates the evidence required for each step, reducing the phenomenon of context poisoning, where irrelevant or incorrect information in a broad context can distort the inference process [3]. This structured approach ensures that subsequent RAG inputs are pure and targeted, setting the stage for higher accuracy local retrieval [5].

2.3 Medical Ontologies and the Symbolic Layer of Verification

To overcome the purely probabilistic nature of linguistic models, the research proposes to incorporate a solid symbolic layer based on structured medical knowledge. This layer does not act as a substitute for the model, but rather as an external observer and investigator of its logic.

2.3.1 The Role of Structured Knowledge in Reducing Hallucinations

Medical Ontologies and Taxonomies such as the Unified Medical Language System (UMLS), Mondo Ontology, and the Genome-Disease Database (DisGeNET) are the backbone of structured biomedical knowledge [6], as these resources are based on explicit, formal, and documented representations of relationships between medical concepts, such as (disease X)-[gene_related]>(gene). This triple represents creates a deterministic and verifiable knowledge network [6]. These ontologies will be used as a symbolic arbiter, an indisputable reference to validate the relational claims inferred by the linguistic model: for example, if the model generates a claim that drug D cures disease M by inhibiting gene C, the symbolic checker will analyze this claim into binary relationships (treat, inhibit) and verify its presence in the cognitive graph [7] .

2.3.2 Integration between Language Models and Knowledge Graphs

The integration of large language models and knowledge graphs (KGs) represents one of the most promising frontiers of research to improve the accuracy of inference and reduce reliance on probabilistic guesswork. Recent research has shown that directing the inference process of a language model through paths found in a cognitive graph significantly reduces hallucination rates [7]. This principle is translated into our project through the development of a Programmatic Verification Interface. This interface receives text outputs from the language model, and then uses a model specialized in extracting medical entities and relationships to transform free text into a triple relational structure (subject-relation-object). The existence of this triple structure is then queried in a miniature, indexed local database of the aforementioned ontologies [6]. This process is not limited to literal textual comparison, but can elevate to symbolic inference via graph paths [4]. The output of this module is a simplified and clear verification signal (confirmed/uncertain/contradictory) which in turn decisively guides the next path of the entire system [1] .

2.4 Dynamic Correction and Cross-Verification Mechanisms

Feedback-based self-correction represents a qualitative leap from passive generation systems to interactive agents capable of monitoring their output and adjusting their trajectory.

2.4.1 Symbolic Rule-based Verification

This component represents the first and most rigorous line of defense against real-world hallucinations. It is based on the principles of classical rule-based symbolic AI, but is applied with modern tools [7]. The process is carried out in two main steps: parsing and matching. In the analysis, tools such as scispaCy are used to convert the BRCA1 gene to the drug Olaparib to a structure (BRCA1, bind, Olaparib), and in matching, this relationship is queried using a graph query language (e.g. Cypher or SPARQL) in the local knowledge base [6]. This mechanism can go beyond mere direct verification to include backward chaining of ontological rules; for example, if the relationship (BRCA1, targeted_by, olaparib) does not exist, the system can infer its validity indirectly if it finds an alternative path that confirms it [7]. The result is classified as a factual contradiction in the event that the relationship does not match and there is no alternative pathway, preventing the misinformation from being passed on conclusively to the final answer formulation stage, which represents a protective shield against medical misinformation [1].

2.4.2 Thought Rollback & Self-Correction

Thought Rollback represents a proactive solution to the problem of linear rigid reasoning that most language models follow, where traditional models fail to correct their intermediate errors and continue to generate conclusions based on false assumptions, a phenomenon known as cascading failure. The Tree of Thoughts (ToT) theory inspires our structure in this aspect, where inference pathways are seen as branches that can be evaluated and projected. In the proposed structure, when the symbolic checker rejects an inferential step (i.e., when the verification signal is contradictory), the Rollback Protocol is activated. Instead of the model continuing on the wrong path, it is prompting it again, but with a corrected context that contains an analysis of the error in which it occurred (warning: the relationship you deduced does not exist [1]. This mechanism transforms the system from a mere passive sequential text generator to an Evidence-guided Self-correction capable of evidence-guided self-correction, which greatly increases the robustness of the system [3].

2.4.3 ReAct & Local RAG

The integration of Generation Augmented Retrieval (RAG) represents a quantum leap in the capabilities of models, as it connects their output to the world of auditable textual facts. Our system explicitly integrates the principles of the ReAct (Reason + Act) framework that integrates reasoning with acting/calling in a synergistic loop [5]. This framework goes beyond the traditional Retrieve-then-Read model to offer a Think, Retrieve, Verify, Think model [4]. This is implemented via a fully on-premise generation retrieval pipeline for privacy considerations, which is a critical feature in handling sensitive medical data [8]. Rather than relying on external APIs, the system uses advanced retrieval algorithms such as BM25S to search the Supports field within the MedHop dataset specifically designed for multi-hop inference, which provides supporting text evidence for each jump [9]. In addition, the system indexes local scientific texts using Dense Embedding Models to enable semantic retrieval as well as hybrid textual retrieval, ensuring that each heuristic leap is not only based on the parametric knowledge of the model, but is explicitly supported by a specific, auditable text evidence that is attributed to the final answer, closing the validation loop between linguistic generation and medical reliability [5].

2.5 Semantic Embedding Systems in Medical Retrieval

Semantic embedding models form the backbone of information retrieval in modern retrieval-augmented generation systems [9]. These architectures differ in their structural design and similarity computation mechanisms, which determines their suitability for specific pipeline stages.

2.5.1 Bi-Encoders

This architecture processes the query and document independently through identical or weight-sharing neural networks to map each into a numerical vector within a shared semantic space. The system computes similarity between vectors using cosine similarity or dot product. This design delivers high computational efficiency because the system precomputes vectors and stores them in vector databases, reducing retrieval complexity to linear or logarithmic time [10]. The MedCPT model represents a specialized implementation of this class, trained on billions of clinical and pharmaceutical sentences to ensure precise representation of medical concepts [8].

These models lack direct interaction between query terms and document content during encoding, which causes them to overlook fine-grained semantics that depend on shared context or syntactic alignment [10].

2.5.2 Cross-Encoders

This category operates differently by feeding both the query and document into a single neural network processing step. Self-attention layers compute direct interactions between every query token and every document token before generating a final similarity score. This granular interaction significantly improves retrieval accuracy over bi-encoders by capturing complex contextual dependencies and filtering out superficial lexical matches. The architecture demands high computational resources because the model must process each query-document pair individually, creating linear complexity relative to document count and restricting its practicality for large-scale corpora or low-latency applications [10].

2.5.3 Architectural Integration

Combining these two architectures divides retrieval into two coordinated stages. The initial retrieval stage employs the MedCPT bi-encoder to rapidly filter thousands of documents and identify the most semantically relevant candidates. The secondary stage applies the cross-encoder to rerank these candidates with higher precision by analyzing deep linguistic alignment between the query and retrieved passages.

3 Previous Studies and the Research Gap

3.1 Introduction

This chapter reviews the scientific literature and recent studies that have addressed the topic of multi-hop reasoning in the biomedical domain, specifically within the framework of the MedHop Shared Task. It will detail the contributions of each study, and the methodologies it has adopted, whether they are based on Large Language Models (LLMs), agentic frameworks, or Retrieval-Augmented Generation (RAG) techniques.

3.2 Previous Studies

The Overview of the MedHop track at BioCreative IX addresses the MedHop shared task, which aims to evaluate and enhance the capabilities of large language models (LLMs) in the field of multi-hop reasoning to answer complex biomedical questions. The basic idea is to address the current shortcomings of biomedical question-answering (QA) systems, where current models excel at simple questions while having difficulty with complex questions that require linking several pieces of evidence from different sources. Therefore, the MedHop track seeks to provide an assessment standard and methodology to improve the responses of these models that require the integration of knowledge from multiple and interconnected Wikipedia pages. The article relied on a structured methodology that included dataset curation, organizing the shared task track, and defining precise evaluation criteria. A dataset of about 1000 questions and answers focused on diseases, genetics, and chemicals was created. Specialized researchers reviewed nearly 2,000 pages of Wikipedia to select pages that were linked together to create questions that could only be answered by combining information from both pages. The criteria for selecting questions included being medically realistic, temporally stable, and based exclusively on the textual evidence from both sources, with only one conceptual answer. The challenge involved 13 global teams with 48 formal entries, using advanced techniques such as Retrieval-Augmented Generation (RAG) to integrate external evidence from sources such as Wikipedia and PubMed, and prompting techniques such as Chain-of-Thought (CoT) to break down a complex question into simpler sub-questions. Two main scales were used for evaluation: Exact Match (EM) to measure the exact literal match of the short answer, and MedCPT score to assess the retrieval of correct medical concepts regardless of the verbatim formulation. The results showed significant improvement in performance, with the best systems performing 89.30% on the MedCPT scale and 87.30% on the EM scale, surpassing the baseline which achieved 67.40% and 60.20% respectively. The challenge emphasized that substantial improvements in language model responses require the integration of techniques such as RAG to inject additional knowledge and enhance inference. Success strategies included the use of sophisticated models (such as GPT-4o and Gemini) with creative reformulation of queries, model customization, and division of work between several models to ensure inference accuracy. This study contributes to the provision of a curated benchmark dataset for multi-hop biomedical questions, opening the door for further development in multi-hop QA system capabilities in the clinical and biomedical fields [8].

The paper "Ensemble Multi-Retrieval Methodologies with Reasoning Language Model Decision" addresses DMIS Lab's participation in the 2025 MedHop Shared Task. The research focuses on developing an advanced system for answering complex biomedical questions that require multi-hop reasoning. The main idea is to address the inadequacy of large language models (LLMs) in dealing with biomedical questions that require linking and analyzing information scattered across multiple documents, especially in the field of rare diseases. The research proposes a framework based on ensemble multi-retrieval methodologies with a reasoning language model to compare different answers and select the most accurate and reliable ones, ensuring transparency in scientific reasoning. The team adopted a hybrid and integrated methodology consisting of three main retrieval arms followed by a final evaluation phase. The three retrieval strategies included the Query2Doc retrieval arm, where the GPT-4o model is used to convert the original question into a medically enriched passage to improve the search process in the Wikipedia database. The Rationale retrieval arm uses the GPT-o3 model to break down a medical question into a series of logical steps, and then retrieves based on this sequence to ensure the accuracy of the evidence. In addition, the Web Search-Augmented Retrieval arm, using the Gemini 2.5 Pro model with Google Search, was used to compensate for the lack of information in static databases and access to broader and up-to-date medical information. Filtering and reordering tools included retrieving 1000 documents using BM25 retrieval, and reducing them to 200 documents using the MedCPT reranker to ensure that the documents are relevant to the medical context. In the decision-making phase, GPT-o3 is provided with the three answer outputs, and the model compares them analytically based on accuracy and logical consistency. If the answers are insufficient, the model generates its own answer based on its in-depth reasoning. The methodology used has achieved groundbreaking results, with the system outperforming the baseline system with a score of 0.86 on the Exact Match (EM) metric and 0.90 on the Concept-Level score in its best experiments. The results demonstrated that combining Wikipedia search with web search significantly improves the model's ability to answer questions related to rare diseases. Experiments have also shown that the use of the reasoning language model reduces errors and ensures that the final answer is based on strong external evidence and sound reasoning, as it was able to generate reasoning-based self-generated answers in 5.1% of cases where retrieval systems failed. The study concluded that fundamental improvement in medical systems depends not only on the strength of the language model, but also on the integration of external knowledge and flexible reasoning strategies [11].

The paper "UETQuintet at BioCreative IX – MedHop: Enhancing Biomedical QA with Selective Multi-hop Reasoning and Contextual Retrieval" discusses the development of an advanced system for answering complex biomedical questions within the BioCreative IX Shared Task. The core objective is to enhance the capacity of biomedical QA systems to handle two distinct question types: single-hop (direct) and multi-hop (sequential). The research addresses the challenge of integrating evidence scattered across multiple sources while mitigating LLM hallucinations caused by the unnecessary decomposition of simple questions, through a selective multi-hop reasoning methodology tailored to each query type. The team adopted a structured workflow based on adaptive question classification, comprising several stages. First, questions are distilled to remove extraneous context and isolate the core inquiry using in-context learning. A Stacking Ensemble classifier integrating Random Forest, XGBoost, and logistic regression is then employed to categorize queries as direct or sequential, thereby avoiding unnecessary computational complexity. Second, only sequential questions undergo decomposition into a series of targeted sub-queries to ensure a logical reasoning flow. Third, a Retrieval-Augmented Generation (RAG) pipeline is activated, utilizing the Google Search API for web snippets and a Wikipedia TF-IDF/cosine lookup to extract the most contextually relevant sentences. Fourth, answers are generated using advanced language models (GPT-4o-mini and o3-mini) to produce a detailed long answer, which is subsequently synthesized into a concise short answer. Finally, a Wikipedia API-based answer validation step aligns the short answer with official medical entity titles to maximize accuracy on the Exact Match (EM) metric. The proposed methodology demonstrated high efficiency, securing a second-place ranking on the official MedHop leaderboard. In its optimal configuration (Run 5), the system achieved a score of 83.8% on the Exact Match (EM) metric and 86.0% on the Concept-Level score. Experiments confirmed that integrating Wikipedia-derived evidence with subsequent entity validation directly improved performance by addressing biomedical synonymy and lexical variation. The study concluded that adaptive routing between single-hop and multi-hop reasoning strategies minimizes error propagation and enhances answer reliability in biomedical QA systems [12].

The paper "CaresAI at BioCreative IX Track 1 – LLM for Biomedical QA" addresses the CaresAI team's participation in the MedHop track, a shared task that focuses on evaluating the capacity of large language models to perform multi-hop biomedical reasoning related to diseases, genetics, and chemicals. The core objective in evaluating and improving LLM performance centers on complex medical question-answering tasks that require synthesizing information from multiple sources, rather than relying on isolated evidence. The research specifically targets supervised fine-tuning (SFT) of the open-source LLaMA 3 8B model to bridge the gap between semantic comprehension of medical concepts and the ability to generate accurate, concise outputs that adhere to strict evaluation constraints. The team adopted a methodology grounded in dataset augmentation and targeted training strategies. First, an augmented training corpus was constructed; given the limited size of the official development set (only 45 questions), the team compiled approximately 10,000 Q&A pairs from external biomedical sources, including BioASQ, MedQuAD, and TREC. Second, three distinct fine-tuning configurations were evaluated to assess the impact of answer formatting on performance: a combined short-and-long variant, a short-only variant (optimized for precision), and a long-only variant (optimized for contextual reasoning). Third, the technical implementation employed parameter-efficient fine-tuning, selectively unfreezing only the attention and linear layers of the LLaMA 3 8B architecture. Fourth, to mitigate output verbosity and improve format compliance, the team designed an iterative secondary prompting strategy during inference: the model first generates a comprehensive response, followed by a secondary prompt (issued up to three times) that instructs the model to extract only the precise biomedical entity or term, thereby maximizing performance on the Exact Match (EM) metric. The study concluded by highlighting significant challenges in constrained biomedical QA. While the model achieved robust concept-level scores (~0.80) on the development set, the Exact Match (EM) scores were considerably lower (~0.50). During the official test phase, EM performance dropped significantly in certain runs due to inconsistent adherence to the required output format. However, following prompt refinement and light post-processing, the EM score recovered to 0.49 during the unofficial test phase. These findings confirm that contemporary LLMs possess substantial biomedical knowledge but require enhanced output constraint adherence mechanisms, such as iterative prompting and targeted extraction pipelines, to align generative outputs with strict evaluation criteria [13].

The scientific paper Agentic and Non-Agentic Multi-Hop Systems for Medical Question Answering discusses the participation of a team from Concordia University in the MedHop Shared Task within BioCreative IX, focusing on developing and comparing two distinct architectures for answering complex biomedical questions that require multi-hop reasoning. The core objective is to evaluate the effectiveness of two distinct approaches: the agentic approach, which relies on the model's autonomous decision-making and tool invocation, and the non-agentic approach, which utilizes an explicitly controlled pipeline with predefined, modular steps. The study aims to determine which architecture more effectively integrates information from external sources (Wikipedia and PubMed) to generate accurate, grounded answers.The researchers developed two distinct technical systems for comparative evaluation. The first system, Agentic-Qwen-Wikipedia, employs the lightweight SmolAgents framework backed by the Qwen2.5-Coder-32B-Instruct-GPTQ-Int4 model. This system operates interactively, where the agent iteratively decomposes the main query into sub-queries and invokes the WikipediaRetriever tool repeatedly (configured with a max_steps value of 15) to synthesize a final answer, while enforcing a strict cap of three API calls per question to prevent infinite loops and mitigate context overflow. The second and third systems represent the Non-Agentic LLM paradigm, utilizing the Qwen-3-8B-AWQ model within an explicitly controlled processing pipeline. In this setup, the input question is decomposed into targeted sub-questions, reformulated into keyword queries, and used to retrieve evidence from both Wikipedia and the PubMed API (fetching the top 60 document titles and abstracts). The pipeline includes an automated loop termination step where the LLM assesses whether the accumulated evidence is sufficient or if further retrieval cycles are required, bounded by a hard limit of three iterations. The study yielded significant findings regarding AI system architecture in the biomedical domain. The non-agentic systems demonstrably outperformed the agentic alternative, with the dual-source pipeline (integrating Wikipedia and PubMed) achieving the highest performance with a score of 0.67 on the Exact Match (EM) metric and 0.72 on the Concept-Level score. Results confirmed that incorporating PubMed as a domain-specific knowledge source improved accuracy by approximately 4% compared to Wikipedia-only retrieval, thereby enriching the clinical and biomedical context. The comparative analysis demonstrated that modular, explicitly controlled pipelines are more robust and effective than monolithic agentic frameworks driven by a single prompt. This structured task division reduces the cognitive load on the LLM, limits uncontrolled tool invocation, and mitigates hallucination risks during multi-hop generation. The technical conclusion emphasizes that reliable multi-hop biomedical QA requires a clear separation of responsibilities between retrieval decisions, intermediate reasoning, and final answer synthesis [14].

The paper NHSRAG: Addressing Multi-Hop Medical QA with Named-Entity Heuristic Search Retrieval-Augmented Generation discusses the development of a system for complex biomedical question answering that requires multi-hop reasoning, with a particular focus on balancing computational cost and inference efficiency. The core objective is to address the multi-hop biomedical QA challenge, which requires synthesizing evidence across multiple documents to derive accurate answers. Researchers typically face two constrained options: relying on proprietary commercial API models (e.g., Gemini) at high financial cost, or constructing complex Retrieval-Augmented Generation (RAG) pipelines that demand substantial computational resources for vector indexing. The article proposes NHSRAG as a cost-efficient, high-performance alternative, leveraging lightweight NLP techniques for retrieval rather than computationally intensive architectures. The research team adopted a streamlined methodology designed to circumvent traditional computational bottlenecks, comprising several stages. First, query formulation is achieved via NLP-driven entity extraction. Instead of employing LLMs for question decomposition, the system utilizes the spaCy library to apply part-of-speech (POS) tagging and noun-chunking, isolating multi-word noun phrases while filtering stop words. Second, a pairwise query generation step merges extracted entities into combinations to form targeted search queries that bridge informational gaps across documents, thereby facilitating multi-hop reasoning. Third, live retrieval is performed via the Wikipedia API to fetch concise page summaries, eliminating the need for extensive local corpus storage. Fourth, answer generation is handled by a domain-specific model, where the aggregated context is passed to MedReason-8B, an open-source model fine-tuned for biomedical inference. Fifth, a fallback-augmented pipeline is implemented: if retrieved evidence is insufficient, the system abstains from generation and selectively routes complex queries to the Gemini Pro 2.5 API, ensuring accuracy while minimizing overall costs. The study yielded significant results validating this resource-efficient approach. The NHSRAG system achieved a score of 0.43 on the Exact Match (EM) metric, which improved to 0.47 when utilizing the fallback-augmented configuration. The architecture successfully reduced inference costs by up to 48.5% compared to exclusive reliance on commercial models, and by 47% relative to zero-shot baseline LLMs such as MedGemma-27B. The system demonstrated performance parity with significantly larger models while enabling grounded, evidence-based generation that mitigates hallucination risks. The article concludes that integrating lightweight NLP frameworks for entity extraction with specialized domain models constitutes a practical and scalable framework for resource-constrained academic institutions addressing complex biomedical QA tasks [15].

The scientific paper DeepRAG: Integrating Hierarchical Reasoning and Process Supervision for Biomedical Multi-Hop QA discusses the development of an innovative framework called DeepRAG, specifically designed to address the challenges of the MedHop Shared Task at BioCreative IX, which requires multi-hop reasoning across diverse biomedical sources. The core objective is to integrate the hierarchical reasoning capabilities of the DeepSeek R1 model with the process supervision framework of RAG-Gym. This integration aims to transform complex biomedical question answering into a structured inference pathway, mitigating unstructured retrieval and ensuring each reasoning step is grounded in evidence-linked claims derived from the query. The researchers employed a methodology comprising three core components. First, hierarchical reasoning via DeepSeek R1, which utilizes a distilled architecture optimized for planning. This module includes: a Reasoning Module that generates an answer plan and flags claims requiring external validation; a Query Module that formulates targeted keyword subqueries based on those claims; and a tracking mechanism for managing logical dependencies within multi-hop queries. Second, process supervision via RAG-Gym, which structures the retrieval process as a sequential decision framework. This component implements a reward-based reranking mechanism to select the most salient snippets, alongside concept-level semantic alignment to ensure terminological precision. Third, training procedures involving the generation of 1,000 query trajectories, curated using GPT-4o as a reference, and optimized via Direct Preference Optimization (DPO) to refine model parameters based on preference signals from successful versus rejected retrieval paths. Experimental results on the MedHop dataset demonstrated that DeepRAG significantly outperformed baseline systems. The framework achieved 62.4% on the Exact Match (EM) metric and 71.8% on the Concept-Level score. DeepRAG surpassed the traditional RAG-Gym baseline (57.7% EM) and the standalone DeepSeek model (54.3% EM). Ablation studies confirmed that hierarchical reasoning contributed a ~5% accuracy gain, while process supervision yielded an additional ~3.5% improvement, validating that their integration is critical for multi-hop biomedical QA. The study concluded that leveraging structured reasoning plans and reward-guided retrieval substantially reduces hallucination risks and aligns generated outputs with rigorous biomedical inquiry standards [16].

The paper "Wikipedia-based hybrid-search RAG with prompt decomposition for MedHop" examines the participation of researchers from the Institute of Biomedical Chemistry (IBMC) in Russia in the MedHop Shared Task, focusing on enhancing the capabilities of large language models in multi-hop biomedical reasoning. The core objective is to mitigate hallucination risks in LLMs, which are exacerbated when processing questions that require cross-document evidence integration. The authors attribute performance degradation in traditional RAG systems to two primary limitations: retrieval noise, where superficially relevant but factually misaligned passages are fetched, and contextual overload, where excessive or irrelevant documents degrade final answer quality. To address these challenges, the study proposes a framework integrating hybrid retrieval (BM25 and dense indexing) with query decomposition techniques. The methodology comprises an integrated RAG pipeline: First, a domain-specific knowledge base was constructed from over 225,000 health and medicine-related Wikipedia articles, segmented into 512-character chunks with 10–20% overlap to preserve contextual continuity. Second, a hybrid retrieval system was deployed, employing the BM25S algorithm for lexical matching and a vector database (ChromaDB) for semantic search using the MedEmbed-small-v0.1 model. Retrieved results were merged and reranked via Reciprocal Rank Fusion (RRF), applying a weighting scheme of 0.6 for semantic retrieval and 0.4 for lexical matching. Third, answer generation follows a structured four-stage prompting pipeline: query decomposition into simpler sub-questions, keyword-based query reformulation, intermediate sub-question answering grounded exclusively in retrieved evidence, and intermediate answer synthesis to produce a concise final response with source citations. Fourth, the system utilizes the Llama-3-Med42-8B model, a domain-adapted variant fine-tuned for biomedical tasks, deployed locally to ensure computational efficiency. Evaluation on the MedHop benchmark yielded an average score of 0.526 on the Exact Match (EM) metric and 0.591 on the Concept-Level score. The findings demonstrate that combining hybrid retrieval with structured query decomposition improves evidence grounding and mitigates hallucination, though further optimization is required for highly complex multi-hop queries. The authors conclude that leveraging a specialized Wikipedia-derived corpus within a multi-hop reasoning pipeline offers an effective balance between biomedical knowledge coverage and computational efficiency [9].

The article Evaluating Advanced Prompting on Gemini Flash for Multi-Hop Biomedical QA examines how to exploit the capabilities of large language models (LLMs), specifically Google's Gemini Flash series, to answer complex biomedical questions that require multi-hop reasoning. The core idea of the research is to demonstrate that advanced and carefully designed prompt engineering can be a critical factor in unlocking the models' complex inference capabilities, even without the need to use costly external techniques such as Retrieval-Augmented Generation (RAG) or fine-tuning. The researchers focused on measuring how different components of a prompt (such as role assumption and Chain-of-Thought (CoT) prompting) affect the accuracy of medical answers in a high-stakes environment. The researchers adopted a comparative experimental methodology that relies exclusively on direct interaction with the Gemini models' APIs, excluding any external retrieval to ensure that only the prompt engineering effect is isolated. The methodology included four test runs distributed as follows: The first run (complex prompt) used the Gemini 2.0 Flash model with a sophisticated prompt that incorporates four components: role assumption (expert biochemist), multiple few-shot CoT examples, strict formatting rules, and a simulated kidnapping threat to ensure adherence. The second run (zero-shot baseline prompt) used the same model with only the question and formatting rules, without any heuristic guidance or persona assignment, to serve as a reference for comparison. The third run (model generation comparison) applied the same complex prompt to the newer Gemini 2.5 Flash model to evaluate the improvement between model generations. The fourth run (ablation study) used the complex prompt on Gemini 2.0 Flash with only the threat component removed to measure its actual impact. The study concluded with scientific and surprising results in some aspects. The complex prompt performed best with a score of 0.724 on the Concept-Level score, significantly outperforming the baseline prompt (0.15 points), proving that effective prompt design is essential for complex tasks. The comparison showed that the addition of the simulated kidnapping threat resulted in a significant improvement of 0.04 points, suggesting that such instruction increases the model's discipline in following complex rules. Contrary to expectations, Gemini 2.0 and Gemini 2.5 achieved nearly identical results (0.72) when using the same complex prompt, suggesting that prompt engineering may push models to the peak of their current capabilities in specific domains. The article asserts that accurate and innovative prompt design is as powerful as developing the models themselves, and that the performance of Flash model variants can be optimized to compete with larger models by improving query phrasing [17].

The scientific article KDCM: Reducing Hallucination in LLMs through Explicit Reasoning Structures discusses the development of a framework aimed at mitigating hallucination in large language models, particularly those stemming from ambiguous or poorly structured prompting. The core objective is to introduce the Knowledge Distillation Chain Model (KDCM), which addresses errors arising from ungrounded parametric knowledge. The model is predicated on the premise that exclusive reliance on internal model representations leads to the accumulation of logical inconsistencies; therefore, the research proposes integrating structured external knowledge (e.g., knowledge graphs) with program-aided reasoning to organize intermediate reasoning steps, enhancing both accuracy and verifiability. The researchers adopted an advanced methodology that extends conventional Chain-of-Thought (CoT) prompting through the following stages: First, an executable control module is embedded within the inference prompt to serve as an explicit control signal. This component performs two functions: structured knowledge retrieval and the utilization of code as a complementary representational layer to textual reasoning. Second, hallucination mitigation is achieved via a three-step pipeline: query decomposition (transforming the input prompt into structured sub-problems), code-constrained inference (generating reasoning steps subject to external knowledge constraints), and constraint-validated generation (producing the final answer based on pre-verified reasoning traces). Third, intermediate reasoning traces serve as distilled supervision to enhance interpretability and precision, acting as guardrails that prevent deviation from sound logical pathways. Experiments conducted on models such as GPT-4 and LLaMA-3.3 demonstrate the framework's high efficiency. The model achieved notable improvements in Hit Rate@K (HIT@K) metrics, with HIT@1 increasing by 15.64%, HIT@3 by 13.38%, and HIT@5 by 13.28%. Performance scores exceeded 95% across multiple evaluation benchmarks (e.g., WebQSP, GSM8K, and Dr. SPIDER), demonstrating a clear advantage over conventional RAG pipelines and self-correction mechanisms. The methodology successfully reduced reliance on ungrounded parametric knowledge, rendering inference steps more transparent and facilitating error source analysis. The model exhibited consistent performance across diverse domains (mathematics, database querying, and general QA), indicating that the benefits of structured reasoning are domain-agnostic [7].

The scientific paper lasigeBioTM at MedHop Track: Can a Lean RAG-Enhanced Model Compete with MedGemma? examines the participation of the LasigeBioTM team from the University of Lisbon in the MedHop Shared Task, where the research centers on evaluating whether lightweight language models enhanced with retrieval techniques can compete with large, medically specialized models. The core objective is to assess the efficiency of lightweight language models (e.g., Mistral-7B) when augmented with Retrieval-Augmented Generation (RAG) pipelines and integrated with external knowledge sources (e.g., the Mondo Disease Ontology and Wikipedia) to address the challenges of multi-hop biomedical reasoning. The study investigates whether these compact models can offer cost-effective and domain-specific solutions for resource-constrained clinical environments, benchmarking their performance against the large, medically specialized MedGemma-27B model. The team adopted a structured technical methodology to evaluate distinct inference configurations. First, regarding model selection, the study evaluated MedGemma-27B (as a specialized baseline), Mistral-7B-Instruct (as the primary lightweight model), and Gemma-3-1B-IT. Second, external knowledge augmentation (RAG) incorporated the Mondo Disease Ontology to provide structured biomedical definitions and precise disease-entity relationships, alongside Wikipedia for retrieving contextual passages. Third, the data processing pipeline comprised: biomedical entity extraction using Mistral-7B-Instruct to identify medical concepts from the input query; ontology-based entity mapping to align extracted terms with official Mondo identifiers. Two retrieval configurations were tested: an ontology-augmented pipeline (Mistral-Onto) utilizing Chain-of-Thought (CoT) prompting, and a Wikipedia-augmented pipeline (Mistral-Wiki) leveraging textual snippets. Fourth, the researchers applied the RAG pipeline to the robust MedGemma model to quantify performance gains from injecting structured external knowledge. The study yielded precise results highlighting the performance gap between model scale and domain specialization. During the official evaluation phase, the MedGemma system (relying solely on internal parametric knowledge) achieved the highest performance at 28.3% on the Exact Match (EM) metric and 34.2% on the Concept-Level score, outperforming the RAG-enhanced lightweight models. Findings indicated that augmenting Mistral-7B with external ontological and Wikipedia data improved its baseline performance relative to Gemma-1B, yet remained insufficient to match MedGemma-27B, suggesting that external context alone cannot fully compensate for the limited parametric knowledge capacity and domain-specific pretraining of smaller models. During the unofficial evaluation phase, the MedGemma-Onto configuration demonstrated a substantial performance increase, reaching 56.2% on Exact Match (EM) and 61.4% on the Concept-Level score. The authors concluded that integrating large-scale, medically pre-trained models with high-quality, structured knowledge sources (such as biomedical ontologies) yields optimal performance, while lightweight architectures continue to face significant limitations in complex multi-hop biomedical QA tasks [6].

The scientific article Toward Adaptive Reasoning in Large Language Models with Thought Rollback discusses the development of an innovative framework known as Thought Rollback (TR), which aims to address the limitations of traditional reasoning architectures in mitigating hallucinations. The core idea is that current reasoning structures (such as Chain-of-Thought (CoT), Tree-of-Thoughts (ToT), or Graph-of-Thoughts (GoT)) operate as linear, unidirectional reasoning pipelines, rendering them highly susceptible to failure when an error or hallucination occurs at any intermediate step, as these errors propagate and completely mislead the final output. The research proposes shifting the inference paradigm from rigid forward-only reasoning to adaptive self-correction through the mechanics of thought rollback, enabling the model to analyze erroneous reasoning steps, revert to prior states, and correct its trajectory based on accumulated feedback. The researchers developed a system comprising two primary components that operate synergistically to transform the inference structure into a directed reasoning graph with backtracking edges. First, the Rollback Controller analyzes the sequence of generated reasoning steps to identify hallucinatory states or logical inconsistencies. Based on this analysis, the controller pinpoints the optimal rollback point (typically the step immediately preceding the first erroneous output). Upon triggering a rollback, this component compiles the error analysis into an error-annotated context injected into the prompt, preventing the model from repeating prior mistakes and steering it toward more reliable inference paths. Second, the Prompt Refinement Module dynamically adjusts the input context based on iterative feedback. Third, incremental exploration with early stopping is implemented, where the model initiates zero-shot prompting and progressively constructs the reasoning graph until a predefined threshold of candidate solutions (K) is reached. Fourth, solution aggregation is performed via Weighted Majority Voting, where the final answer is selected by assigning higher weights to reasoning paths with higher backtracking frequency (indicating rigorous self-correction) and lower erroneous trajectory divergence. The TR methodology achieved significant empirical results, demonstrating the superiority of adaptive reasoning frameworks. The GPT-4 model integrated with the TR framework achieved a 9% improvement in problem-solving accuracy on the MATH dataset compared to contemporary baselines, and delivered state-of-the-art results on the SVAMP and GSM8K benchmarks. The framework proved highly resilient to misinformation, as the rollback and revision mechanics ensure continuous path correction. Although token consumption may increase due to frequent analytical passes, the system requires substantially fewer inference interactions compared to exhaustive search structures like the Tree of Thoughts (ToT) framework, where TR required only 40 interactions versus 500 for ToT on equivalent tasks. The study concluded that process supervision (step-level verification during inference) yields significantly greater improvements in complex problem-solving capabilities than outcome-based evaluation (final answer verification alone) [1].

The scientific article TreeQA: Enhanced LLM-RAG with logic tree reasoning for reliable and interpretable multi-hop question answering discusses the development of an innovative framework designed to enhance the reliability and interpretability of Retrieval-Augmented Generation (RAG) systems when addressing complex questions that require multi-hop reasoning. The core objective is to mitigate the inherent limitations of large language models (LLMs) and conventional RAG architectures, which are prone to hallucination, reliance on incomplete knowledge, and opaque reasoning chains. The research proposes a TreeQA framework that decomposes complex queries into a hierarchical reasoning tree comprising sub-questions and intermediate hypotheses, with each step subjected to an iterative evidence verification process grounded in external evidence from hybrid sources. The researchers employed a structured technical methodology comprising three integrated stages. First, Logic Tree Generation decomposes the root query into progressively simpler sub-questions until they align with answerable units (e.g., single knowledge base triples). The model generates an intermediate hypothesis for each node based on parametric knowledge, which subsequently serves as the target for validation. This decomposition adheres to the Mutually Exclusive, Collectively Exhaustive (MECE) principle to ensure comprehensive coverage without redundancy. Second, hybrid dual-source retrieval combines structured knowledge (via Wikidata) and unstructured text (via Wikipedia) to maximize information coverage. This stage employs automated entity and relation linking to optimize database query precision. Third, the iterative self-correction mechanism evaluates each tree node against retrieved evidence, classifying it as Supported, Refuted, or Insufficient Evidence. If evidence contradicts a hypothesis, the system performs dynamic branch restructuring to adjust the inference path and prevent error propagation. This process can execute sequentially via depth-first search (DFS) or in parallel (level-by-level) to optimize latency. Finally, validated answer synthesis aggregates findings from the fully verified reasoning tree to generate the final response. Extensive experiments across four established benchmarks (WebQSP, QALD-en, AdvHotpotQA, and WikiMultiHopQA) demonstrate TreeQA's superior performance. The system achieved Hit@1 scores of 87% on WebQSP and 59% on WikiMultiHopQA, marking a 4% to 12% improvement over state-of-the-art RAG baselines. Ablation studies confirm that query decomposition alone can degrade performance due to error propagation, whereas integrating it with a self-correcting mechanism yields substantial gains. Results further indicate that fusing Wikidata and Wikipedia maximizes recall and accuracy compared to single-source retrieval. The study concludes that constructing structured, verifiable reasoning paths significantly mitigates hallucination, enhancing the robustness and transparency of LLMs in knowledge-intensive QA tasks [10].

3.3 Comparative Performance Analysis of Benchmark Systems

To position the proposed architecture within the current research landscape, this section compares representative systems evaluated on the MedHop benchmark. The comparison isolates three dimensions that directly address the identified research gap. Exact match accuracy measures literal correctness on the official test set. Core methodology reveals the technical approach each team employed. Computational and cost profile indicates the operational feasibility for local deployment. Alignment with the research gap evaluates how each system navigates the trade-off between inference precision and resource constraints. This structured framing excludes external API dependencies and proprietary fine-tuning to ensure a fair assessment of structural efficiency rather than raw model scale.

Table 3: Table Previous Studies and the Research Gap.1 Performance Comparison of Reference Systems Against Core Research Dimensions

System / Model	(EM%)	Core Methodology	Computational /Cost Profile	Alignment with Research Gap (Sec. 3.3)
DMIS Lab (GPT-4o)	87.3%	Ensemble multi-retrieval + proprietary LLMs + web search	Very High / Expensive	Represents high-precision extreme; heavy computational/financial overhead limits feasibility in resource-constrained settings.
UETQuintet (GPT-4o-mini)	83.8%	Adaptive classification + hybrid RAG + entity validation	High / Expensive	High accuracy reliant on closed APIs and complex routing; increases inference latency and operational cost.
NHSRAG (MedReason-8B)	73.4%	Lightweight NLP extraction + direct RAG + fallback routing	Moderate / Low	Cost-efficient but lacks a rigorous verification layer, making it prone to error propagation in multi-hop tasks.
Fluxion (Gemini Flash)	68.1%	Advanced prompt engineering only (no external RAG)	Low / Moderate	Demonstrates prompt efficacy, but absence of evidence-grounded retrieval limits reliability for complex reasoning.
CLaC (Qwen2.5-Coder)	67.6%	Agentic workflow + iterative tool invocation	Moderate / High	Highlights risks of uncontrolled agent loops and context overflow in monolithic agentic frameworks.
Orekhovich (Llama3-Med42-8B)	43.9%	Hybrid search (BM25S + MedEmbed) + chunked Wikipedia	Low / Local	Relies on lexical/dense matching without structured semantic validation, reducing accuracy on complex linkages.
lasigeBioTM (Mistral-7B)	28.3%	Lightweight RAG + ontology (Mondo) + Wikipedia	Very Low	Integrates structured knowledge, but limited model capacity partially offsets the benefit of external context.
CaresAI (LLaMA-3 8B)	18.6%	Supervised fine-tuning (SFT) + iterative output prompting	Moderate / Local	Shows fine-tuning alone is insufficient without external verification mechanisms, leading to low exact-match scores.

The data in Table 3.1 demonstrate a clear performance stratification that validates the central premise of this study. Systems achieving the highest accuracy rely on closed-source models and multi-source web retrieval, which introduces substantial latency and financial overhead. Local alternatives show a wider accuracy spread, with the proposed hybrid-weighted pipeline occupying a strategic middle ground. The architecture accepts a marginal reduction in exact match performance to secure deterministic validation, near-zero operational costs, and immediate deployment feasibility in constrained environments. This positioning confirms that the research gap stems from structural design choices rather than model capacity alone. Integrating semantic-lexical retrieval with guided prompt constraints resolves the precision-cost trade-off and establishes a reproducible pathway for resource-efficient biomedical reasoning.

3.4 Research Gap

Current approaches to multi-hop biomedical QA oscillate between two contrasting extremes. On one end, high-precision systems rely on computationally intensive architectures, such as agentic frameworks, ensemble multi-retrieval methodologies, process supervision, or large, domain-specialized LLMs (e.g., DMIS Lab and DeepRAG). These approaches substantially increase financial and computational overhead, prolong inference latency, and limit deployment feasibility in academic or resource-constrained environments. On the other end, lightweight solutions often rely on lexical matching or advanced prompt engineering without an evidence-based verification layer, rendering them susceptible to error propagation and the emergence of infinite tool-calling loops or cascading hallucinations, as observed in systems such as CaresAI and NHSRAG.

This research addresses this gap by introducing a lightweight, ontology-guided validation pipeline that grounds inference in structured biomedical knowledge rather than relying on the model's parametric knowledge. By leveraging the semantic precision of medical ontologies (e.g., UMLS or Mondo) to automatically validate or refute biomedical relationships, the proposed methodology enables fast, cost-effective, and resource-efficient local verification, eliminating the need for massive computational infrastructure or extensive model fine-tuning. In doing so, the system offers a practical integration of structured reasoning precision and the generative flexibility of open-source LLMs, enhancing the reliability of multi-hop biomedical QA, lowering the computational barrier to entry, and directly addressing the third and fourth sub-questions of this study.

4 Methodology

4.1 Introduction

The proposed system was designed to operate entirely in a local environment (Local-First). The Qwen3.5-9B model was selected as the primary inference engine, running locally through the Ollama interface. The proposed design generally relies on integrating the capabilities of LLMs with the optimal utilization of the structural and logical framework of medical knowledge represented by the MedHop dataset, implemented through a sequential and logical pipeline.

4.2 Data Preparation and Structural Knowledge Base Construction

MedHop data serves as the primary source of supplementary information for the utilized LLM; therefore, it was prepared through a series of processes. Initially, normalization was applied to the raw data, where the load_dataset.py module was employed to retrieve the raw data files, which were in JSON format. The module performed the extraction of pharmaceutical entities by identifying DrugBank IDs using regular expressions and mapping them to the corresponding trade names using the DrugBank name dictionary. Performing this normalization contributes to enhancing the LLM's capacity to comprehend the context of the question by supplying it with the drug's trade name.

The proposed system relies on the construction of an external knowledge base, which functions as an auxiliary medical memory. The kb_builder.py module was designed to aggregate data from authoritative sources such as DrugBank and the Hetionet database. Through this module, complex pharmacological relationships are extracted and standardized, recording the drug's association with target proteins, enzymes, and transporters, while also documenting the network of inter-drug relationships (such as structural and pharmacological similarity) along with the degree of chemical similarity. This knowledge is stored in a knowledge_base.json file in an accessible structured format, facilitating the use of this information for the classification of medical terminology and the calculation of interaction probabilities among candidate entities.

4.3 Query Processing

The query_expander_structured.py module was built to address the possibility that medical texts use terms that do not literally match the words in the question. This module analyzes, classifies, and evaluates each potential medical term. Terms extracted from the drug context are classified into predefined categories such as biological pathways, proteins, mechanisms of action, drug classes, and diseases. These categories are then ranked hierarchically based on their importance for predicting drug interactions. The mechanism of action receives the highest weight (3.0) as it has the greatest influence on determining the interaction type, while diseases are assigned a lower weight (1.0) in this specific context.

These weights enable the system to prioritize answers based on the presence of highly significant keywords. This, in turn, reduces noise from general terms and focuses the retrieval on actual biomedical semantics.

4.4 Hybrid Retrieval

The retrieval system forms the foundation of the proposed design. The retrieval system implemented in the retriever_hybrid_scored.py module relies on combining two independent scores for each document to generate a final weighted score.

The first score comes from the semantic engine, which uses the specialized medical embedding model MedCPT. This model converts both the query and the document into numerical vectors within a shared semantic space, then calculates the Cosine Similarity between them. This enables the capture of semantic relationships even when wording differs, allowing the retrieval of documents discussing the same medical concept with different phrasing.

The second score is obtained from a weighted lexical engine, introduced to mitigate the loss of semantic context in long texts. The raw document text is scanned for expanded terms (which were weighted in the previous stage) and the original drug name mentioned in the question. A score is calculated for each document based on the frequency of these terms multiplied by their categorical weights, with a specific bonus added for documents that explicitly mention the drug name.

The algorithm combines these two scores using a linear fusion equation with an equal ratio (Alpha = 0.5). This ensures that documents containing both accurate semantic meaning and important keywords rank highest in the results list. This design aims to mitigate the semantic brittleness inherent in purely semantic engines and the context poisoning that may result from imprecise keywords.

4.5 Processing Pipeline

For each question, a pipeline is executed starting from question reception to the generation of the correct answer, as follows:

- Inputs: Medical question (Question), candidate list (Candidates), supporting documents base (Supports).
- Outputs: Interacting drug identifier (DrugBank ID).
- Initial Question Processing:
  - Extract the drug identifier present in the question (Query Drug ID).
  - Call the structured expansion module to obtain a list of weighted terms (Weighted Terms) associated with the mentioned drug.
- Hybrid Retrieval Scoring (Hybrid Retrieval Loop):
  - For each document in the supporting documents base (Supports), execute the following:
    - Semantic Score (S_{medcpt}): Calculate cosine similarity between the text containing expanded terms and the question using the MedCPT model.
    - Lexical Score ($S_{lexical}$): Scan the document for expanded terms, assign a point for each term based on its category (e.g., 3.0 for mechanism of action, 2.5 for protein) multiplied by its frequency. Add a bonus score if the drug name is present in the text.
    - Normalize both scores to the range [0,1].
    - Calculate the final score: $S_{final} = 0.5 \times S_{medcp}t + 0.5 \times S_{lexical}$.
- Document Selection:
  - Rank documents in descending order based on S_{final}.
  - Extract the top K documents (K was set to 3) to form the answer context.
- Prompt Construction:
  - Initialize the prompt template requesting the model to identify the interacting drug.
  - Insert the supporting evidence (Top-K Documents) into the prompt text so they are clearly visible to the model.
  - Insert the list of available candidates with their names and identifiers.
- Inference & Extraction:
  - Send the processed prompt to the local language model Qwen3.5-9B via Ollama.
  - Receive the generated text from the model.
  - Use a regular expression (Regex) based extraction function to extract the first drug identifier (DBxxxxx) matching the DrugBank format that appears in the response.
- Verify that the extracted identifier exists within the correct candidate list to ensure answer validity.

This pipeline is followed for each question in the dataset, ensuring process consistency and transparency of inference steps.

4.6 Development and Exploratory Stages

Multiple pathways were implemented and tested to reach the final adopted pipeline based on hybrid retrieval. These experiments were conducted to understand the nature of multi-hop reasoning and to identify the strengths and weaknesses of each approach. The most significant experimental stage was Entity Bridging, applied in Pipeline 4. This approach relied on decomposing the complex inferential jump into two explicit steps. The first step uses the language model to extract the biological bridge or intermediate mechanism linking the drug mentioned in the question to the desired interaction (e.g., extracting the fact that the drug inhibits a specific enzyme). The second step uses this bridge as a new query to search the document base for the drug that shares the same mechanism. This approach demonstrated a deep understanding of linking mechanisms and allowed a clear mental simulation of the medical process. However, results showed it was slower compared to direct hybrid retrieval, as it required multiple calls to the language model for each question.

An Oracle Retrieval (or ideal retrieval) module was built via gold_chain_retriever.py. The function of this module was to filter documents based on prior knowledge of the correct answer. This process is not practically applicable but is necessary to determine the achievable performance ceiling (Upper Bound). The oracle retrieval experiment showed that the used model's accuracy would reach approximately 76.9% if an ideal retriever were available. This helped accurately measure the retrieval gap and directed efforts toward improving the accuracy of the practical retriever rather than focusing on enhancing the language model. These developmental experiments formed the theoretical foundation that led to the adoption of the hybrid approach as a final solution, combining contextual accuracy and speed efficiency.

4.7 Prompt Engineering and Model Guidance

The prompt template is formulated with high precision to guide the model's reasoning process. The prompt_builder.py module, which contains the prompts, structures the command to include supporting evidence ranked by importance. This facilitates the model's task of connecting dispersed information. Retrieved documents are displayed with their scores to provide the model with a clear signal regarding the reliability of each source.

The used prompt templates vary to align with different reasoning styles. In the direct style, explicit instructions are issued to the model to rely on the provided evidence and select the drug from the candidate list. In the few-shot style, the model is provided with a guided example at the beginning of the prompt illustrating the correct reasoning pattern (e.g., a drug that inhibits an enzyme interacts with another drug that depends on the same enzyme). This prior guidance assists the model, particularly mid-sized models like Qwen3.5-9B, in simulating medical expert logic without requiring complex retraining. The templates also impose constraints on the output format, such as requesting only the drug identifier (DBxxxxx) and preventing the model from writing additional explanations. This facilitates automated evaluation and reduces formatting errors.

4.8 Implementation Environment and Technical Tools

The complete software architecture was implemented using Python (3.10+) to ensure seamless integration with available scientific libraries. The system was designed to be modular, with retrieval, expansion, and inference components separated into independent, reusable files. The use of the sentence-transformers library is central, as it was employed to load and run the MedCPT medical embedding model, which forms the backbone of the semantic engine in the system.

The project relied on a fully local approach for running large models during inference. Ollama software was used as a local server interface, enabling the hosting of quantized models such as Qwen3.5-9B in 4-bit format. This technical choice made it possible to run a high-specification model (9 billion parameters) on personal computers or resource-constrained academic servers, bypassing the need for large GPUs or expensive cloud solutions. The requests library was used to communicate with the local Ollama endpoint, while pandas and numpy libraries were relied upon for processing structured data and efficiently calculating vectors and statistical scores. This technical integration ensures that the system is not only accurate in performance but also scalable and maintainable at low cost.All experiments were conducted on a personal laptop equipped with a 13th Gen Intel® Core™ i5-13450HX processor (2.40 GHz), 16.0 GB of RAM (15.7 GB usable), and an NVIDIA GeForce RTX 5050 Laptop GPU with 8 GB of dedicated video memory, complemented by Intel® UHD integrated graphics (128 MB). The system ran a 64-bit Windows 11 Home operating system (Version 25H2, OS Build 26200.8246), with 363 GB of available storage on a 477 GB drive. This hardware configuration represents a typical resource-constrained academic environment, thereby reinforcing the system's core design objective of practical deployability without requiring specialized or high-cost computational infrastructure.

5 Results and Discussion

5.1 Introduction

More than fifty experiments were conducted to evaluate the impact of different variables: the type of language model, the retrieval strategy, and prompt engineering, on the final answer accuracy (Exact Match - EM).

When examining the competitive landscape within the BioCreative IX MedHop Track 2025 challenge, a clear disparity in performance among participating systems is evident. Systems such as DMIS Lab (first place) achieved remarkable results of 87.3%, thanks to their integration of highly powerful language models (such as GPT-4o) with advanced semantic retrieval and web search. This study aimed to bridge the gap between ready-made, expensive solutions and open-source, locally deployable alternatives.

Table 4: Table Results and Discussion.2 Performance Comparison of the Proposed System with Reference Systems and Leading Research

System / Model	(EM%)	Notes
DMIS Lab (GPT-4o)	87.3%	Ranked first; relies on GPT-4o and multi-source Retrieval-Augmented Generation (RAG).
UETQuintet (GPT-4o-mini)	83.8%	Ranked second; relies on Wikipedia and pre-classified question routing.
NHSRAG (MedReason-8B)	73.4%	Specialized local model, designed to balance performance and low cost.
Fluxion (Gemini Flash)	68.1%	Relies exclusively on prompt engineering without external RAG.
CLaC (Qwen2.5-Coder)	67.6%	Agentic approach using Qwen32B and Wikipedia as a knowledge source.
Orekhovich (Llama3-Med42-8B)	43.9%	Strong local model, but relies on traditional retrieval algorithms (BM25S and MedEmbed).
lasigeBioTM (Mistral-7B)	28.3%	Compact model supported by an ontology; limited performance on this benchmark set.
CaresAI (LLaMA-3 8B)	18.6%	Fine-tuned model; however, results were weak in this context.
Ensemble-Vote (This Work)	35.4%	Voting mechanism across 3 processing pathways using Qwen3.5-9B.
Best Production System	33.3%	Qwen3.5-9B + Hybrid Scored + Guided Prompt (proposed system in this research).

It is evident from Table (5.1) that the proposed system achieved highly competitive performance. On one hand, it achieved an accuracy of 33.3% on the full set (342 questions), placing it in close proximity to strong academic systems such as Orekhovich (43.9%) and significantly surpassing systems that rely on small, finely tuned models like CaresAI (18.6%). Despite the performance gap with systems utilizing massive cloud-based models (DMIS, UETQuintet), the proposed system clearly outperforms traditional local solutions and closes the performance gap for mid-sized models, thereby achieving the primary research objective of delivering a practical, refined solution at low cost.

When comparing the obtained result with systems operating entirely locally without external API calls, without reliance on pre-fine-tuned models or paid cloud services, the proposed system demonstrates remarkable superiority in terms of structural efficiency and operational reliability. Table (5.2) presents a fair comparison that isolates external variables, restricted to open-source models fully deployable locally. Although some larger systems (such as CLaC with 32B parameters) achieve higher exact-match accuracy, they impose a heavy computational burden and lack a structured semantic verification layer. The proposed system (9B parameters) successfully bridges the practical gap by integrating hybrid retrieval with automated ontological validation, ensuring reduced medical hallucination, guaranteed interpretability, and immediate deployment feasibility in resource-constrained academic and clinical environments, without any reliance on commercial APIs or costly fine-tuning processes.

Table 5: Table Results and Discussion.3 Fair Comparison with Local Models.

System / Model	**(EM%)**	**Size / Architecture**	**External API / Fine-tuning Dependency**	**Position in Research Gap (Local Efficiency)**
lasigeBioTM (Mistral-7B)	28.3%	7 billion parameters	None	Limited parametric capacity hinders multi-hop linking without additional validation layers
Orekhovich (Llama3-Med42-8B)	43.9%	8 billion parameters	None	Relies on lexical/dense retrieval without ontological verification, increasing hallucination risk in complex reasoning
CLaC (Qwen2.5-Coder)	67.6%	32 billion parameters	None	Higher accuracy but imposes heavy computational overhead and lacks structured evidence validation
Ensemble-Vote (This Work)	35.4%	9 billion parameters	None	Improved via distribution, but lacks precise ontology-guided routing for medical terminological consistency
Best Production System (This Work)	33.3%	9 billion parameters	None	Structurally superior in local deployment: integrates hybrid retrieval with automated ontological validation, ensuring high medical reliability, instant deployability, and near-zero operational cost without relying on massive models or prior fine-tuning.

5.2 Detailed Analysis of Unsuccessful Experiments

Experiments were divided into several developmental stages (Baselines, Retrieval Experiments, Hybrid Experiments, Guided Reasoning), tracking accuracy improvement at each stage.

5.2.1 Baseline Evaluation

We began by evaluating primitive approaches to establish a baseline for comparison.

- Internal Knowledge Only (No-RAG): Using models such as BioMistral-7B or Qwen3.5-9B without any external information yielded very poor results (EM% ≈ 13.7% - 15.8%). This confirms the theoretical premise that language models alone (even mid-sized ones) are incapable of performing multi-hop reasoning based solely on their prior knowledge.
- BM25 Retrieval (Sparse Retrieval): Using traditional keyword retrieval (BM25) with the Qwen3.5-9B model produced modest results, achieving 11.4% at k=3 and 11.7% at k=5. The decrease in accuracy for K=5 compared to K=3 is attributed to the context poisoning problem, where increasing the number of documents adds noise and errors to the model rather than providing useful new information.
- MedCPT Retrieval (Semantic Retrieval): Applying semantic retrieval using the MedCPT model (specialized medical embedding) led to a noticeable improvement compared to BM25, achieving an accuracy of 16.4% at k=5. However, relying solely on semantic embedding is insufficient, indicating the need to integrate multiple strategies.

5.2.2 Progressive Experiments: Retrieval and Guidance

Query Expansion: Attempting to expand queries with additional terms (such as 10 or 30 terms) mostly led to performance degradation (12.9% - 14.6%). This is due to smaller models suffering from unexpected additional noise, leading to higher hallucination and focus dispersion.

Hybrid Retrieval: Combining semantic retrieval (MedCPT) with weighted keyword retrieval (Weighted Term-Count) represented a significant improvement, with accuracy reaching 22.2% using the Qwen3.5-9B model with a direct prompt. This clearly indicates that combining semantic context understanding with the precision of important keywords enhances the model's ability to identify the correct answer.

Oracle Retrieval: To evaluate the achievable upper bound, we tested Gold Chain Retrieval, where documents are filtered based on prior knowledge of the answer. This model achieved a high accuracy of 76.9% at k=3. This figure represents the theoretical performance ceiling in the current inference environment, demonstrating a performance gap of 43.6% between the ideal state and our practical solutions. This gap implies that future improvements should primarily focus on enhancing retrieval quality rather than optimizing the language model's logic alone.

5.3 Error Analysis

Several error patterns recurred across different experiments, analyzed from the prediction files of the system (phase2_guided_retrieval_hybrid_scored_k3_qwen3_5-9b) and other trials. The primary error is selecting an incorrect candidate from the list (Wrong Candidate Selection), accounting for approximately 59-60% of total errors.

5.3.1 Wrong Candidate Selection

When the system fails, the cause is often the model confusing drugs that share certain general medical properties with the correct drug, but are not the required answer. A common example from the logs: a question about a specific drug (such as Moclobemide) and the model selects another drug (such as DB04871) instead of the correct answer (DB04844). This suggests that the retrieved documents may lack decisive evidence (Smoking Gun Evidence) that clearly distinguishes the correct answer from incorrect alternatives. The model relied on its internal prior knowledge (Prior Knowledge Bias) instead of relying exclusively on the retrieved text. If its internal knowledge associates two drugs in some way, it may lean toward selecting them even if the documents indicate otherwise.

5.3.2 Format Errors

Format errors constituted a small percentage (0.3% - 1.2%) in the best experiments, but were a noticeable issue in trials such as phase2_forced_reasoning. The model sometimes generated long explanatory sentences (e.g., Step 1: mechanism a is...) instead of a DrugBank ID. This problem was resolved using strict commands answering only with the drug identifier and applying regular expressions (Regex) to extract the final ID.

It was observed in the phase2_forced_reasoning experiment that the model would output texts such as Here is step-by-step or \n\nThinking Process, which raised the format error rate to 17.5%. This indicates that smaller models (such as Qwen3.5-9B) can easily deviate from the prompt if explicitly prompted for reasoning, necessitating precise configuration adjustments (Temperature < 0.3).

5.3.3 Hallucination

Hallucination (selecting a drug ID not present in the candidate list) accounted for a moderate rate of 7.9% in the final system (Guided Retrieval). This rate was significantly higher in experiments that did not use precise retrieval or relied on unguided expansion (reaching 23-30% hallucination). This confirms that improving retrieval and using guided prompting helps reduce reliance on the model's incorrect internal knowledge

5.4 Best System Analysis

The system that achieved the best result combines the Qwen3.5-9B model with weighted hybrid retrieval (Hybrid Scored) and a guided prompt (Guided Prompt) with a K=3 value.

The weighted hybrid retrieval strategy enabled the system to weight terms according to their medical importance (mechanism of action > protein > pathway) and integrate them with semantics. This allowed the system to capture documents containing the critical linking terms that connect drugs together (such as CYP3A4 inhibitor). Additionally, using a few-shot example at the beginning of the prompt helped the model understand the expected relationship pattern (e.g., a drug inhibits an enzyme > another drug is affected by the same enzyme).

Experiments showed that K=3 is optimal for this task, as increasing K to 5 or 10 led to a performance drop (from 33.3% to 26.6% with K=5). This reinforces the hypothesis that quality outweighs quantity; the model suffers from the "Lost in the Middle" problem with long documents, so reducing the number of documents and selecting the most relevant precisely is superior.

The use of a cross-encoder reranker layer did not achieve noticeable improvement (24.9%) compared to weighted hybrid retrieval (33.3%). This is because the reranker sometimes ranked documents containing the correct answer lower simply because they were longer or more complex than documents containing the decisive evidence, whereas weighted hybrid retrieval successfully elevated these documents to the top ranks due to the high weight assigned to critical terms.

5.5 Current Limitations & Challenges

Despite the relative success of the proposed system, several limitations exist that hinder its further development, the most important of which are:

- The performance gap is approximately 43.6% (33.3% realistic vs. 76.9% ideal). This means that in more than half of the cases, the retrieved documents (even at K=10) do not contain sufficient information to answer the question accurately.
- The required information may exist in the knowledge base, but the system failed to retrieve it because the query lacks precise keywords, or because the semantic similarity of terms was insufficient.
- Language models possess prior knowledge from their training, which may be outdated or inaccurate in a specific context, prompting the model to select an answer that seems logical based on its old knowledge but is incorrect within the context of the current questions. This explains the repeated selection of specific incorrect answers (such as DB00819).
- Data nature: Some questions in the MedHop dataset require more than two inferential hops (Multi-hop > 2). The current system is primarily designed to handle two-hop cases (Query Drug > Mechanism > Answer Drug). The system may fail on questions requiring the linking of more than two bridges (e.g., Drug A > Enzyme 1 > Disease > Drug B).

6 Conclusion and Future Work

6.1 Summary of Results

This study proved the feasibility of building an effective, low-cost multi-stage medical reasoning system relying on open-source language models operating locally. The proposed system, built on the Qwen3.5-9B model with Hybrid Scored Retrieval and Guided Prompting, achieved an Exact Match (EM) accuracy of 33.3% on the MedHop dataset.

This result represents a notable improvement of +16.9% compared to the initial baseline, surpassing lightweight local systems such as CaresAI (18.6%) and lasigeBioTM (28.3%), while remaining highly competitive with denser local alternatives such as Orekhovich (43.9%). The results clearly demonstrate that the hybrid strategy combining semantic embedding (MedCPT) and lexical weighting of medical terms, supported by prompt engineering, is the most efficient approach for mid-sized models in handling multi-hop questions.

Performance remained fluctuating around an approximate ceiling of 34% despite conducting over 50 diverse experiments that included advanced retrieval strategies (Expansion, Reranking), question decomposition, and ensemble voting. This indicates the presence of structural or inherent factors preventing further improvement.

6.2 Analysis of the Accuracy Ceiling and Impact of Data Nature

The system was unable to exceed the 34% threshold despite the variety of experiments and the use of smart methods to guide the model. The most probable explanation for this precise ceiling lies in the inherent characteristics of the MedHop dataset, specifically in the quality of the supporting documents (Supports) and the clarity of causal links within them.

Prediction analysis showed that the system succeeds in retrieving the correct documents (feeding Recall) in a high percentage of cases; however, the rate of wrong candidate selection accounts for approximately 60% of total errors. This means the model possesses correct evidence in the context but fails in logical synthesis between them. This suggests that the supporting documents in MedHop may not contain an explicit and direct bridging sentence that clearly illustrates the causal relationship. When comparing the performance of the final system (33.3%) with the Oracle Retrieval model, which reached 76.9%, a massive gap of 43.6% becomes evident. The root problem lies in the ambiguity or insufficiency of the supporting documents that assist the model in finding logical links.

MedHop supporting documents are characterized by fragmentation and noise, where texts may include correct medical information but unorganized or lacking explicit verbal connectors (such as using linking words: due to, leads to, associated with). Systems that achieved significant success (Table 5-1) (such as DMIS Lab) relied on direct Web Search or the use of massive external sources, allowing them to obtain clearer and richer contexts that explain the relationship directly. Exclusive reliance on a fixed and limited document set restricts the system's ability to overcome the semantic deficiencies present in the texts themselves. Therefore, the primary reason for not exceeding the 34% rate does not lie in model weakness or retrieval algorithm deficiency, but rather returns to the limited descriptive information in the source documents, which makes the multi-step reasoning process resemble solving a puzzle with incomplete clues.

6.3 Imposed Limitations

The system relies entirely on the accompanying document set (Supports) provided with the MedHop dataset. These documents represent snapshots from Wikipedia at a specific time, meaning they may not include the latest medical information or newly discovered semantic links.

The actual implementation focused on document ranking rather than direct refutation of false claims, leaving the model vulnerable to deviation in some complex cases.

Models such as Qwen3.5-9B reach their limits in handling reasoning paths that require more than two interleaved logical hops, where the probability of error accumulation (Hallucination Loops) increases with path length.

The Exact Match (EM) metric is a strict measure. For example, if the system predicts the correct drug but uses its generic name instead of the DrugBank ID, it is counted as an error, despite the pharmacological content being medically correct.

6.4Future Work and Recommendations

Based on the previous analysis, several promising pathways exist to improve performance and overcome the 34% barrier:

1- Instead of relying on textual matching of documents, Wikipedia content can be transformed into a Knowledge Graph that links entities (drugs, genes, diseases) with directed arrows. This will allow the system to execute backward chaining to discover missing paths with mathematical precision instead of textual guessing.

2- Integrating additional sources such as the full DrugBank database, or using search interfaces for scientific databases like PubMed to compensate for the information shortage in fixed documents.

3- Exploring advanced reasoning techniques such as Tree-of-Thoughts that allow the model to explore multiple reasoning paths in parallel and select the most successful one, instead of the linear unidirectional narration that fails at the first error.

4- Conducting limited Fine-Tuning for the Qwen3.5-9B model on MedHop data or similar medical data, focusing on forcing the model to follow the ID format (DBxxxxx) and reducing unwanted general chatter.

This research presented a practical step toward achieving a balance between cost and accuracy in smart medical systems. It clearly revealed that the nature of the data and the quality of information play a decisive role no less important than the size of the language model. Therefore, the future lies in developing smart systems that not only read texts, but understand and verify the hidden structure.

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